

A Posteriori Error Analysis

Adaptive Finite Element Methods for Elliptic PDEs — Lecture 5

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The problem adaptivity must solve

- **Computable information:** Adaptivity needs **computable** error information. We build an estimator $\mathcal{E}_{\mathcal{T}}(U)$ depending only on the data (f, A, Ω) and the computed Galerkin solution $U \in \mathbb{V}_{\mathcal{T}}$.
- **Two-sided equivalence:** **Two-sided bound** between the true energy error and estimator

$$\underbrace{C_2 \mathcal{E}_{\mathcal{T}}^2(U) - \text{osc}_{\mathcal{T}}^2(U)}_{\text{efficiency}} \leq |u - U|_{H^1(\Omega)}^2 \leq \underbrace{C_1 \mathcal{E}_{\mathcal{T}}^2(U)}_{\text{reliability}}.$$

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- **Goals of this lecture:** We will prove
 - **Reliability** (upper bound): $\mathcal{E}_{\mathcal{T}}(U)$ small $\Rightarrow |u - U|_{H^1(\Omega)}$ small. Prevents *under*-refinement.
 - **Efficiency** (lower bound, up to *oscillation*): $\mathcal{E}_{\mathcal{T}}(U)$ large $\Rightarrow |u - U|_{H^1(\Omega)}$ large. Prevents *over*-refinement.
- **Routine MARK:** The **local indicators** $\{\mathcal{E}_{\mathcal{T}}(U, T)\}_{T \in \mathcal{T}}$ are exactly what MARK uses in Lecture 6.

Roadmap

- 1 Model problem, residual, and the error–residual relation
- 2 Residual representation and Galerkin orthogonality
- 3 Reliability: the upper bound
- 4 Efficiency: the lower bound
- 5 Oscillation and the full lower bound
- 6 Consequences
- 7 The adaptive loop

Model problem, residual, and the error–residual relation

Model problem and residual

- **Model problem:**

$$\begin{cases} -\operatorname{div}(A(x)\nabla u) = f & \text{in } \Omega \subset \mathbb{R}^d, \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

with weak solution $u \in \mathbb{V} = H_0^1(\Omega)$ and Galerkin solution $U \in \mathbb{V}$ (C^0 piecewise linears).

- **Bilinear form:** $\mathcal{B}[u, v] = \int_{\Omega} A \nabla u \cdot \nabla v$, coercive and bounded w.r.t. the H^1 -seminorm $|v|_{H^1(\Omega)} = \|\nabla v\|_{L^2(\Omega)}$ (a norm on H_0^1 by Poincaré inequality):

$$\mathcal{B}[v, v] \geq \alpha |v|_{H^1(\Omega)}^2, \quad |\mathcal{B}[u, v]| \leq M |u|_{H^1(\Omega)} |v|_{H^1(\Omega)}.$$

- **Residual** $R \in H^{-1}(\Omega)$ **dual of** $H_0^1(\Omega)$: it depends only on (f, A, Ω) and $U \in \mathbb{V}_{\mathcal{T}}$

$$\langle R, v \rangle = \langle f, v \rangle - \mathcal{B}[U, v] = \mathcal{B}[u - U, v] \quad \forall v \in \mathbb{V},$$

$$\|R\|_{H^{-1}(\Omega)} = \sup_{v \in \mathbb{V} \setminus \{0\}} \frac{\langle R, v \rangle}{|v|_{H^1(\Omega)}}.$$

Error–residual relation

- **Equivalence:** Energy error $\simeq$ dual norm of the residual

$$\alpha \|u - U\|_{H^1(\Omega)} \leq \|R\|_{H^{-1}(\Omega)} \leq M \|u - U\|_{H^1(\Omega)}.$$

- **Lower bound:** test against $v = u - U$ and use coercivity of $\mathcal{B}$

$$\alpha \|u - U\|_{H^1(\Omega)}^2 \leq \mathcal{B}[u - U, u - U] = \langle R, u - U \rangle \leq \|R\|_{H^{-1}(\Omega)} \|u - U\|_{H^1(\Omega)},$$

and divide by $\|u - U\|_{H^1(\Omega)}$.

- **Upper bound:** Use boundedness of $\mathcal{B}$

$$|\langle R, v \rangle| = |\mathcal{B}[u - U, v]| \leq M \|u - U\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}.$$

Divide by $\|v\|_{H^1(\Omega)}$ and take the supremum.

Error–residual relation

- **Equivalence:** Energy error $\simeq$ dual norm of the residual

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and divide by $|u - U|_{H^1(\Omega)}$.

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$$|\langle R, v \rangle| = |\mathcal{B}[u - U, v]| \leq M |u - U|_{H^1(\Omega)} |v|_{H^1(\Omega)}.$$

Divide by $|v|_{H^1(\Omega)}$ and take the supremum.

- **Goal:** It remains to bound $\|R\|_{H^{-1}(\Omega)}$ by *computable* quantities.

Residual representation and Galerkin orthogonality

Residual representation: integration by parts

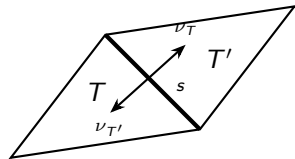
- **Element-by-element integration by parts:** With $A \in C^{0,1}(\bar{\Omega}; \mathcal{T}_0)$ pw Lipschitz, $f \in L^2(\Omega)$,

$$\langle R, v \rangle = \sum_{T \in \mathcal{T}} \int_T f v - \int_T A \nabla U \cdot \nabla v = \sum_{T \in \mathcal{T}} \int_T [f + \operatorname{div}(A \nabla U)] v - \int_{\partial T} A \nabla U \cdot \nu_T v.$$

Interior-edge boundary terms survive only as **flux jumps**.

- **Interior residual:** $r_T := f + \operatorname{div}(A \nabla U)|_T$.
- **Jump residual:** across interior $S = T \cap T'$,

$$j_s := \llbracket A \nabla U \cdot \mathbf{n} \rrbracket_S = -(A_T \nabla U_T \cdot \nu_T + A_{T'} \nabla U_{T'} \cdot \nu_{T'}).$$



$j_s = \llbracket A \nabla U \cdot \nu \rrbracket$: jump of the normal flux.

Residual representation: integration by parts

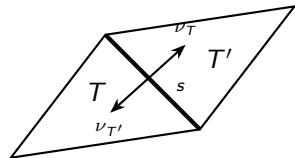
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- **Constant A over $T \in \mathcal{T}$:** For $\mathcal{P}_1$ we have $\operatorname{div}(A \nabla U) = 0$, so $r_T = f|_T$.

Residual representation and Galerkin orthogonality

Residual identity

$$\langle R, v \rangle = \sum_{T \in \mathcal{T}} \int_T r_T v + \sum_{s \in \mathcal{S}} \int_s j_s v, \quad \mathcal{S} = \text{mesh skeleton (set of edges)}.$$

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Galerkin orthogonality — the key trick

Since $\langle R, I_{\mathcal{T}} v \rangle = \mathcal{B}[u - U, I_{\mathcal{T}} v] = 0$ for the interpolant $I_{\mathcal{T}} v \in \mathbb{V}_{\mathcal{T}}$,

$$\langle R, v \rangle = \langle R, v - I_{\mathcal{T}} v \rangle \quad \forall v \in \mathbb{V}.$$

We may subtract $I_{\mathcal{T}} v$ for free; local interpolation estimates then localize the residual to element patches.

Reliability: the upper bound

Reliability

Theorem 1 (reliability)

There is $C = C(\text{shape regularity})$ with

$$|u - U|_{H^1(\Omega)} \leq C \mathcal{E}_{\mathcal{T}}(U), \quad \mathcal{E}_{\mathcal{T}}(U) = \left(\sum_{T \in \mathcal{T}} \mathcal{E}_{\mathcal{T}}^2(U, T) \right)^{1/2},$$

and local indicator

$$\mathcal{E}_{\mathcal{T}}^2(U, T) = h_T^2 \|r_T\|_{L^2(T)}^2 + h_T \|j\|_{L^2(\partial T)}^2.$$

Two proofs with the same estimate:

- ❶ quasi-interpolant $I_{\mathcal{T}}$ (Clément / Scott–Zhang),
- ❷ hat-function ϕ_z localization and partition of unity.

Reliability — Proof 1 (quasi-interpolant $I_{\mathcal{T}}$)

- **Bound for $\|R\|_{H^{-1}(\Omega)}$:** By Galerkin orthogonality and residual representation, it suffices to bound

$$|\langle R, v \rangle| = \left| \sum_T \int_T r_T(v - I_{\mathcal{T}}v) + \sum_{S \in \mathcal{S}} \int_S j_S(v - I_{\mathcal{T}}v) \right| \quad \forall v \in H_0^1(\Omega).$$

- **Cauchy–Schwarz elementwise:**

$$|\langle R, v \rangle| \leq \sum_{T \in \mathcal{T}} \|r_T\|_{L^2(T)} \|v - I_{\mathcal{T}}v\|_{L^2(T)} + \sum_{S \in \mathcal{S}} \|j_S\|_{L^2(S)} \|v - I_{\mathcal{T}}v\|_{L^2(S)}.$$

- **Quasi-interpolation estimates:** (ω_T, ω_s = patches of stars around T, s)

$$\|v - I_{\mathcal{T}}v\|_{L^2(T)} \lesssim h_T \|\nabla v\|_{L^2(\omega_T)}, \quad \|v - I_{\mathcal{T}}v\|_{L^2(s)} \lesssim h_s^{1/2} \|\nabla v\|_{L^2(\omega_s)},$$

the latter via a scaled trace inequality.

Reliability — Proof 1, conclusion

- **Combine over elements/edges:** Substitute and apply Cauchy–Schwarz,

$$|\langle R, v \rangle| \lesssim \underbrace{\left(\sum_T h_T^2 \|r_T\|_{L^2(T)}^2 + \sum_s h_s \|j_s\|_{L^2(s)}^2 \right)^{1/2}}_{= \mathcal{E}_{\mathcal{T}}(U)} \underbrace{\left(\sum_T \|\nabla v\|_{L^2(\omega_T)}^2 \right)^{1/2}}_{\lesssim \|\nabla v\|_{L^2(\Omega)}}.$$

- **Finite overlap:** **Finite overlap** of patches ω_T gives $\lesssim \|\nabla v\|_{L^2(\Omega)} = |v|_{H^1(\Omega)}$, whence

$$\|R\|_{H^{-1}(\Omega)} = \sup_v \frac{|\langle R, v \rangle|}{|v|_{H^1(\Omega)}} \lesssim \mathcal{E}_{\mathcal{T}}(U),$$

and the error–residual relation finishes the proof. □

Reliability — Proof 2 (partition of unity)

- **Partition of unity:** Hat functions $\{\phi_z\}_{z \in \mathcal{N}} \subset \mathbb{V}_{\mathcal{T}}$ form a partition of unity $\sum_{z \in \mathcal{N}} \phi_z = 1$. Galerkin orthogonality gives $\langle R, \phi_z \rangle = 0$ at interior nodes, so

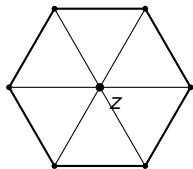
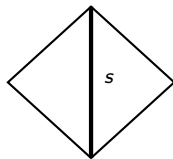
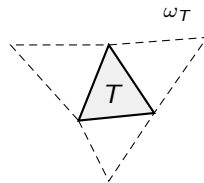
$$\langle R, v \rangle = \sum_z \langle R, v \phi_z \rangle = \sum_z \langle R, (v - \bar{v}_z) \phi_z \rangle, \quad \bar{v}_z = \begin{cases} f_{\omega_z} v, & z \notin \partial\Omega, \\ 0, & z \in \partial\Omega. \end{cases}$$

- **Poincaré inequalities:** On each star ω_z :

$$\|v - \bar{v}_z\|_{L^2(\omega_z)} \lesssim h_z \|\nabla v\|_{L^2(\omega_z)}, \quad \|v - \bar{v}_z\|_{L^2(\partial\omega_z)} \lesssim h_z^{1/2} \|\nabla v\|_{L^2(\omega_z)}.$$

- **Conclusion:** Cauchy–Schwarz over stars yields the same estimate as Proof 1. This proof provides better control of interpolation constants (Verfurth–Veese). □

Local patches


 ω_z (node star)

 ω_s (edge patch)

 ω_T (neighborhood)

Finite overlap of patches ensures the $\|\nabla v\|$ factors recombine globally into $\|\nabla v\|_{L^2(\Omega)}$.

Efficiency: the lower bound

From reliability to efficiency

- **Why efficiency:** Reliability prevents *under*-refinement. For optimality we need the converse: the estimator must not *overestimate* the error — **efficiency**.
- **Strategy: bubble functions.**

Test the residual against localized *bubble* functions supported on one element (interior residual) or one edge patch (jump residual). This inverts the residual representation locally.

- The argument is *exact* when the data (hence r, j) is piecewise constant.
- For general data the price is a higher-order **oscillation** term.

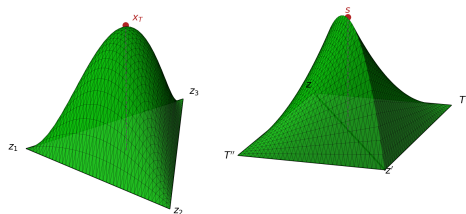
Element-bubble functions

• Cubic bubbles:

Cubic element bubble $b_T = 27\lambda_1\lambda_2\lambda_3$ (barycentric λ_i):

$$b_T > 0 \text{ in } \text{int}(T), \quad b_T = 0 \text{ on } \partial T, \quad |\nabla b_T| \lesssim h_T^{-1},$$

$$\int_T b_T \, dx \simeq |T| \simeq h_T^2, \quad \int_T b_T^2 \, dx \simeq |T| \simeq h_T^2.$$



Element bubble (left) and edge bubble (right).

• Quadratic bubbles: $q_{T,ij} = 4\lambda_i\lambda_j$ for edge $[z_i, z_j] \subset T$

$$\int_T q_{T,ij} \, dx \simeq |T| \simeq h_T^2, \quad \int_T q_{T,ij}^2 \, dx \simeq |T| \simeq h_T^2.$$

• Properties: These integral scalings — not any pointwise normalization — are all the proof uses.

Interior residual: local lower bound (constant r)

- **Test with the element bubble:** Let $r = r_T$ be constant on T . Using b_T as a test function,

$$\begin{aligned}\|r\|_{L^2(T)}^2 &= |T| r^2 \lesssim \int_T b_T r^2 = \langle r, r b_T \rangle \leq \|r\|_{H^{-1}(T)} \|\nabla(r b_T)\|_{L^2(T)} \\ &= \|r\|_{H^{-1}(T)} \|r \nabla b_T\|_{L^2(T)} \lesssim \|r\|_{H^{-1}(T)} h_T^{-1} \|r\|_{L^2(T)}.\end{aligned}$$

- **Local lower bound:** Therefore $h_T \|r\|_{L^2(T)} \lesssim \|r\|_{H^{-1}(T)}$. Since $\langle r_T, v \rangle = \mathcal{B}[u - U, v]$ for $v \in H_0^1(T)$, we have $\|r_T\|_{H^{-1}(T)} \leq \|\nabla(u - U)\|_{L^2(T)}$, giving

$$h_T \|r\|_{L^2(T)} \leq \|\nabla(u - U)\|_{L^2(T)}.$$

Edge-bubble functions

- **Edge bubble:** $b_s = 4\phi_z\phi_{z'}$ (product of the two hat functions of the endpoints of s):

$$0 \leq b_s \leq 1, \quad |\nabla b_s| \lesssim 1/|s|,$$

- **Integral scalings:** for $T \subset \omega_s$ (with $\omega_s = T' \cup T''$ the support),

$$\int_T b_s \, dx \simeq h_T^2, \quad \int_T b_s^2 \, dx \simeq h_T^2, \quad \int_s b_s \, d\sigma \simeq |s| \simeq h_T, \quad \int_s b_s^2 \, d\sigma \simeq |s|.$$

Jump residual: local lower bound (constant j) — I

- **Test with the edge bubble:** For j_s constant on s ,

$$\|j_s\|_{L^2(s)}^2 = |s| j_s^2 \lesssim \int_s j_s^2 b_s = \int_s \llbracket A \nabla(U - u) \cdot \mathbf{n} \rrbracket_s j_s b_s.$$

- **Integrate by parts:** over the support $\omega_s = T' \cup T''$, on each T ,

$$\int_s \llbracket A \nabla(U - u) \cdot \mathbf{n} \rrbracket_s j_s b_s = \sum_{T \in \{T', T''\}} \int_T \operatorname{div}[A \nabla(U - u) j_s b_s].$$

- **Product rule:** then $\operatorname{div}(A \nabla(U - u)) = -r_T$ and Cauchy–Schwarz,

$$\int_T \operatorname{div}[\cdots] = \int_T r_T j_s b_s + A \nabla(U - u) \cdot \nabla(j_s b_s) \lesssim (h_T \|r_T\|_{L^2(T)} + \|\nabla(U - u)\|_{L^2(T)}) |j_s|.$$

Jump residual: local lower bound (constant j) — II

- **Cancel one power:** Since $|j_s| \simeq h_s^{-1/2} \|j_s\|_{L^2(s)}$,

$$\int_s \llbracket A \nabla(U - u) \cdot \mathbf{n} \rrbracket_s j_s b_s \lesssim (h_s \|r\|_{L^2(\omega_s)} + \|\nabla(U - u)\|_{L^2(\omega_s)}) h_s^{-1/2} \|j_s\|_{L^2(s)},$$

so that

$$h_s^{1/2} \|j_s\|_{L^2(s)} \lesssim h_s \|r\|_{L^2(\omega_s)} + \|\nabla(u - U)\|_{L^2(\omega_s)}.$$

- **Local bound (pw constant data):**

$$h_T^2 \|r\|_{L^2(T)}^2 + h_T \|j\|_{L^2(\partial T)}^2 \lesssim |u - U|_{H^1(\omega_T)}^2.$$

Oscillation and the full lower bound

Oscillation: Variable data (f, A)

- **Residual splitting:** The bubble argument is exact only for constant r, j . Split residuals (r_T, j_s) into a constant part $(\bar{r}_T, \bar{j}_s)$ and a remainder:

$$r = \bar{r}_T + (r - \bar{r}_T), \quad \bar{r}_T = \fint_T r, \quad j_s = \bar{j}_s + (j_s - \bar{j}_s), \quad \bar{j}_s = \fint_s j_s.$$

- **Oscillation:** Constant parts obey the equivalence; the remainders give the **local oscillation**

$$\text{osc}_{\mathcal{T}}(U, T)^2 = h_T^2 \|r - \bar{r}_T\|_{L^2(T)}^2 + h_T \|j - \bar{j}\|_{L^2(\partial T)}^2, \quad \text{osc}_{\mathcal{T}}^2(U) = \sum_{T \in \mathcal{T}} \text{osc}_{\mathcal{T}}^2(U, T).$$

- **Properties of oscillation:**

- $\text{osc}_{\mathcal{T}}(U)$ vanishes for piecewise-constant data.
- $\text{osc}_{\mathcal{T}}(U)$ is asymptotically higher-order when f, A are smooth (extra power of h).

Local bound for the interior residual

Lemma 1

$$h_T \|r_T\|_{L^2(T)} \lesssim \|r_T\|_{H^{-1}(T)} + h_T \|r_T - \bar{r}_T\|_{L^2(T)} \quad \forall T \in \mathcal{T}.$$

Proof

- Triangle inequality gives $h_T \|r\|_{L^2(T)} \leq h_T \|\bar{r}_T\|_{L^2(T)} + h_T \|r - \bar{r}_T\|_{L^2(T)}$. Constant-residual equivalence applied to $\bar{r}_T$ yields

$$h_T \|\bar{r}_T\|_{L^2(T)} \lesssim \|\bar{r}_T\|_{H^{-1}(T)} \lesssim \|r\|_{H^{-1}(T)} + \|r - \bar{r}_T\|_{H^{-1}(T)}.$$

- For any $v \in H_0^1(T)$, using $\langle r - \bar{r}_T, v \rangle = \langle r - \bar{r}_T, v - \bar{v}_T \rangle$ and Poincaré inequality implies

$$|\langle r - \bar{r}_T, v \rangle| \leq \|r - \bar{r}_T\|_{L^2(T)} \|v - \bar{v}_T\|_{L^2(T)} \lesssim h_T \|r - \bar{r}_T\|_{L^2(T)} \|\nabla v\|_{L^2(T)},$$

- Hence $\|r - \bar{r}_T\|_{H^{-1}(T)} \lesssim h_T \|r - \bar{r}_T\|_{L^2(T)}$. Combine the estimates.

Local bound for the jump residual

Lemma 2

$$h_s^{1/2} \|j_s\|_{L^2(s)} \lesssim |u - U|_{H^1(\omega_s)} + h_s^{1/2} \|j_s - \bar{j}_s\|_{L^2(s)} + \sum_{T \subset \omega_s} \|r\|_{H^{-1}(T)} \quad \forall s \in \mathcal{S}.$$

- **Idea:** Split $j = \bar{j}_s + (j - \bar{j}_s)$ and bound $h_s^{1/2} \|\bar{j}_s\|_{L^2(s)}$ by testing $\bar{j}_s b_s$ as before:

$$\|\bar{j}_s\|_{L^2(s)}^2 \lesssim \int_s j \bar{j}_s b_s + \int_s \bar{j}_s (j - \bar{j}_s) b_s.$$

- **Terms:** The first term is controlled by the constant-jump computation; the second by Hölder with the oscillation remainder. The $\|r\|_{H^{-1}(T)}$ terms feed back the interior residual on ω_s .

Efficiency (local lower bound)

Theorem 2 (efficiency)

For each $T \in \mathcal{T}$, with neighborhood ω_T ,

$$\mathcal{E}_{\mathcal{T}}(U, T) = \left(h_T^2 \|r\|_{L^2(T)}^2 + h_T \|j\|_{L^2(\partial T)}^2 \right)^{1/2} \lesssim \|\nabla(u - U)\|_{L^2(\omega_T)} + \text{osc}_{\mathcal{T}}(U, \omega_T).$$

Summing over $T \in \mathcal{T}$ yields

$$C_2 \mathcal{E}_{\mathcal{T}}^2(U) \leq \|u - U\|_{H^1(\Omega)}^2 + \text{osc}_{\mathcal{T}}^2(U).$$

Sketch of proof:

- Interior term: $h_T \|r\|_{L^2(T)} \lesssim \|\nabla(u - U)\|_{L^2(T)} + \text{osc}_{\mathcal{T}}(U, \omega_T)$
- Jump residual: $h_s^{1/2} \|j_s\|_{L^2(s)} \lesssim \|\nabla(u - U)\|_{L^2(\omega_s)} + h_s^{1/2} \|j_s - \bar{j}_s\|_{L^2(s)} + \sum_{T \subset \omega_s} \|r\|_{H^{-1}(T)}$
- Sum up and invoke finite overlapping property of the patches ω_T 's

Consequences

Two-sided control — the main result

Estimator $\simeq$ energy error, up to oscillation

$$C_2 \mathcal{E}_{\mathcal{T}}^2(U) - \text{osc}_{\mathcal{T}}^2(U) \leq |u - U|_{H^1(\Omega)}^2 \leq C_1 \mathcal{E}_{\mathcal{T}}^2(U).$$

- Oscillation is asymptotically of **higher-order** (Lecture 9)
- If A is piecewise constant, then

$$\text{osc}_{\mathcal{T}}(U) = \text{osc}_{\mathcal{T}}(f) = \|h(f - \hat{f})\|_{L^2(\Omega)}$$

reduces to **data oscillation**. This reduces with refinement.

- In general, neither $\mathcal{E}_{\mathcal{T}}(U)$ or $\text{osc}_{\mathcal{T}}(U)$ reduce with refinement; they might even increase because they depend on the discrete solution that changes with refinement.
- If $\text{osc}_{\mathcal{T}}(U) \ll \mathcal{E}_{\mathcal{T}}(U)$, then $\mathcal{E}_{\mathcal{T}}(U)$ is a faithful, computable proxy for the energy error.
- Two-sided bound is the foundation for convergence (Lecture 8) and optimality (Lecture 9) results.

Interior node property

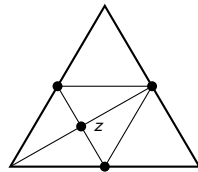
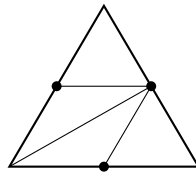
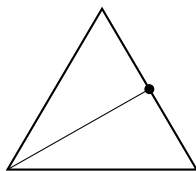
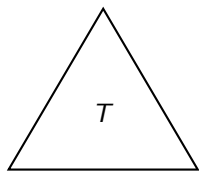
- Interior node as test function:** When $\mathcal{T}_* \geq \mathcal{T}$ adds an interior node $z \in T$, the hat function $\phi_z \in \mathbb{V}(\mathcal{T}_*)$ ($\int_T \phi_z \approx |T|$) replaces the bubble b_T . Testing $\langle r_T, \phi_z \rangle$ and using Galerkin orthogonality with the new solution U_* ,

$$\langle r_T, \phi_z \rangle = \mathcal{B}[u - U, \phi_z] = \mathcal{B}[U_* - U, \phi_z] \Rightarrow h_T \|r_T\|_{L^2(T)} \lesssim \|\nabla(U_* - U)\|_{L^2(\omega_T)} + \text{osc}_{\mathcal{T}}(U, \omega_T).$$

Why it matters

Refining $\mathcal{T}$ (adding an interior node) *directly* reduces the estimator by the discrete energy increment $\|\nabla(U_* - U)\|$ — the link between the marked set and the energy decrement used in the contraction proof of Lecture 8.

Interior node property — refinement



Successive bisections of T (and neighbors, for conformity) until an interior node $z \in T$ appears; ϕ_z then acts as a localized test function.

Counterexample of $\mathcal{E}_{\mathcal{T}}(U) \approx \|\nabla(u - U)\|_{L^2(\Omega)}$

- Fix a mesh $\mathcal{T}$ and a function $f \in H^{-1}(\Omega) \setminus L^2(\Omega)$.
- Let $(f_k)_{k \in \mathbb{N}}$ be a sequence of approximations in $L^2(\Omega)$ of f such that

$$\|f - f_k\|_{H^{-1}(\Omega)} \rightarrow 0.$$

- The exact and Galerkin solutions $u_k \in H_0^1(\Omega)$ and $U_k \in \mathbb{V}_{\mathcal{T}}$ remain bounded because $\|f_k\|_{H^{-1}(\Omega)}$ are uniformly bounded, whence $\|\nabla(u_k - U_k)\|_{L^2(\Omega)}$ remains uniformly bounded.
- But $\mathcal{E}_{\mathcal{T}}(U_n) \rightarrow \infty$ because $\|hf_k\|_{L^2(\Omega)} \rightarrow \infty$ as $k \rightarrow \infty$ with h fixed.

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- But $\mathcal{E}_{\mathcal{T}}(U_n) \rightarrow \infty$ because $\|hf_k\|_{L^2(\Omega)} \rightarrow \infty$ as $k \rightarrow \infty$ with h fixed.
- **Lower bound:** The bound

$$\mathcal{E}_{\mathcal{T}}(U) \lesssim \|\nabla(u - U)\|_{L^2(\Omega)}$$

is generically false!

The adaptive loop

AFEM: The adaptive loop

- Four modules:** The estimator $\mathcal{E}_{\mathcal{T}}(U, T)$ just built is the last ingredient needed to state AFEM itself: a loop over four independent **black-box modules**: only its input and output matter.
 - $U = \text{SOLVE}(\mathcal{T})$: the Galerkin solution on $\mathcal{T}$ (Lecture 3).
 - $\{\mathcal{E}_{\mathcal{T}}(U, T)\}_{T \in \mathcal{T}} = \text{ESTIMATE}(U, \mathcal{T})$: the indicators of this lecture.
 - $\mathcal{M} = \text{MARK}(\{\mathcal{E}_{\mathcal{T}}(U, T)\}, \mathcal{T})$: elements selected for refinement (Dörfler marking, Lecture 6).
 - $\mathcal{T}_* = \text{REFINE}(\mathcal{T}, \mathcal{M})$: smallest conforming refinement bisecting every $T \in \mathcal{M}$ (Lectures 4, 6).
- AFEM:** Given an initial conforming mesh $\mathcal{T}_0$, set $k = 0$ and iterate

$$\begin{aligned}
 U_k &= \text{SOLVE}(\mathcal{T}_k) \\
 \{\mathcal{E}_k(U_k, T)\}_{T \in \mathcal{T}_k} &= \text{ESTIMATE}(U_k, \mathcal{T}_k) \\
 \mathcal{M}_k &= \text{MARK}(\{\mathcal{E}_k(U_k, T)\}_{T \in \mathcal{T}_k}, \mathcal{T}_k) \\
 \mathcal{T}_{k+1} &= \text{REFINE}(\mathcal{T}_k, \mathcal{M}_k); \quad k \leftarrow k + 1
 \end{aligned}$$

Lecture 6 turns each module into a Matlab/Octave function; Lectures 7–9 prove the loop converges, and with optimal rate.

Summary

- ❶ **Error $\simeq$ residual:** $|u - U|_{H^1(\Omega)} \simeq \|R\|_{H^{-1}(\Omega)}.$
- ❷ **Representation:** $\langle R, v \rangle = \sum_T \int_T r_T v + \sum_s \int_s j_s v$, with Galerkin orthogonality

$$\langle R, v \rangle = \langle R, v - I_{\mathcal{T}} v \rangle.$$
- ❸ **Reliability:** $|u - U|_{H^1(\Omega)}^2 \leq C_1 \mathcal{E}_{\mathcal{T}}^2(U)$ (quasi-interpolant or partition of unity).
- ❹ **Efficiency:** $C_2 \mathcal{E}_{\mathcal{T}}^2(U) \leq |u - U|_{H^1(\Omega)}^2 + \text{osc}_{\mathcal{T}}^2(U)$ (bubble functions).
- ❺ **Two-sided bound:** equivalent to error up to oscillation; $\mathcal{E}_{\mathcal{T}}(U)$ drives MARK (Lecture 6).
- ❻ **Interior node property:** discrete lower bound; strict error reduction (Lecture 8).
- ❼ **The adaptive loop AFEM:** this is realized in code in Lecture 6

SOLVE $\rightarrow$ ESTIMATE $\rightarrow$ MARK $\rightarrow$ REFINE.

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SOLVE $\rightarrow$ ESTIMATE $\rightarrow$ MARK $\rightarrow$ REFINE.

Next: Lecture 6 — implementation of the adaptive loop AFEM.